

FINANCIAL MANAGEMENT (FTX2024F)

Solution to Practical Tutorial Question - Advanced Valuation

WORKINGS:

W1

Calculation of investment in NWC:

	Actual 20X7 Rm	Projected 20X8 Rm	Projected 20X9 Rm
Investment in NWC (20% of sales)	200.0	260.0	325.0
Annual change in NWC investment (negative = outflow)		-60.0	-65.0

W2

Calculating FCF to the firm during the explicit forecast period and PV of FCFTF and CV

	Actual 20X7 Rm	Projected 20X8 Rm	Projected 20X9 Rm
Sales	1 000.0	1 300.0	1 625.0
NOPAT (using the EBIT approach):			
EBIT before non-operating income (40% of sales)	-	520.0	650.0
Less: Tax (EBIT x 28%)	-	(145.6)	(182.0)
NOPAT [EBIT x (1 – t)]	-	374.4	468.0
Add: Depreciation	-	80.0	90.0
Gross operating cash flow to the firm	-	454.4	558.0
Less: Net CAPEX investment	-	(100.0)	(120.0)
Less: Investment in NWC	-	(60.0)	(65.0)
Free cash flow to the firm	-	294.4	373.0
CV _{20x9} [(R373m x 1.06)/(0.14 – 0.06)]	-		4 942.3
FREE CASH FLOW TO THE FIRM INCLUDING CV	-	294.4	5 315.3
PVIF (Discounting using WACC of 14% - see W0)	-	0.8772	0.7695
PV OF FREE CASH FLOWS TO FIRM	-	258.2	4 090.1

W3

Calculating the value of the firm and the value of ordinary shares:

	Rm
PV of FCFTF (258.2 + 4 090.1)	4 348.3
ADD: Non-operating assets	100.0
ENTERPRISE (FIRM) VALUE	4 448.3
LESS: Fair value of the loan	(380.0)
VALUE OF ORDINARY SHARES	4 068.3
Value of a 60% shareholding (4 068.3 x 60%)	2 441.0